1 Week 9

Let m be a set function defined for all sets in a σ-algebra $\mathcal{A}$ with values in $[0, \infty]$. Assume m is countably additive over countable disjoint collections of sets in $\mathcal{A}$.

Question 1.1 (Royden 2.1)**.** Prove that if A and B are two sets in $\mathcal{A}$ with $A \subseteq B$, then $m(A) \leq m(B)$. This property is called *monotonicity*.

Proof. Since $A = A \sqcup (B \setminus A)$ we have $m(B) = m(A) + m(B \setminus A)$. From this equality, it is evident that A may not have infinite measure unless B does as well, so the inequality $m(A) \leq m(B)$ is obvious for when B has infinite measure. For B with finite measure, we have $m(A) \leq m(A) + m(B \setminus A) = m(B)$ since $m(B \setminus A) \geq 0$. $\square$

Question 1.2 (Royden 2.2)**.** Prove that if there is a set A in the collection $\mathcal{A}$ for which $m(A) < \infty$ then $m(\varnothing) = 0$.

Proof. Since $\varnothing \subseteq A$, we have by Royden 2.1 that $m(\varnothing) \leq m(A)$ is finite. Since $\varnothing = \varnothing \sqcup \varnothing$, we have $m(\varnothing) = m(\varnothing) + m(\varnothing) \Rightarrow m(\varnothing) = 0$. $\square$

Question 1.3 (Royden 2.3)**.** Let $\{E_k\}_{k=1}^{\infty}$ be a countable collection of sets in $\mathcal{A}$. Prove that $m(\bigcup_{k=1}^{\infty} E_k) \leq \sum_{k=1}^{\infty} m(E_k)$.

Proof. Define the collection of sets $\{F_k\}_{k=1}^{\infty}$ by $F_k = E_k \setminus \bigcup_{i=1}^{k-1} E_i \in \mathcal{A}$. It is clear that $\bigcup_{k=1}^{\infty} F_k \subseteq \bigcup_{k=1}^{\infty} E_k$. For the reverse inclusion, given an element $x \in \bigcup_{k=1}^{\infty} E_k$, we take the smallest natural n for which $x \in E_n$ and note that $x \in F_n$ by construction of the F_k's. Therefore the two unions are equal. Furthermore, given positive integers $n < m$, we have that $F_n \subseteq E_n$ and $F_m \cap E_n = \varnothing$ by construction, so $F_n \cap F_m = \varnothing$. Thus the collection $\{F_k\}_{k=1}^{\infty}$ is a disjoint collection of sets in $\mathcal{A}$ and we have

$$m\left(\bigcup_{k=1}^{\infty} E_k\right) = m\left(\bigcup_{k=1}^{\infty} F_k\right) = \sum_{k=1}^{\infty} m(F_k) \leq \sum_{k=1}^{\infty} m(E_k)$$

as required. $\square$

Question 1.4 (Royden 2.4)**.** A set function c, defined on all subsets of $\mathbb{R}$ is defined as follows. Define $c(E)$ to be ∞ if E has infinitely many members and $c(E)$ to be the number of elements in E if E is finite; define $c(\varnothing) = 0$. Show that c is a countably additive and translation invariant set function. This set function is called the **counting measure**.

Proof. For translation invariance, we simply note that $f : E \to E + y, e \mapsto e + y$ is a bijection for any $E \in \mathcal{P}(\mathbb{R})$ and $y \in \mathbb{R}$. For countable additivity, consider a countable collection $\{E_k\}$ of mutually disjoint subsets of $\mathbb{R}$. Since there are sets of finite cardinality, we know that $c(\varnothing) = 0$ by Royden 2.2, and hence can remove empty sets from the collection without any effect on the union or the sum of measures. We thus can assume that $\{E_k\}$ is a collection of nonempty sets. If it is an infinite collection, then we have $c(E_k) \geq 1$ and hence $\sum_k c(E_k) \geq \sum_k 1 = \infty$. We also have in this case that $\bigcup_k E_k$ contains infinitely many elements, and hence $c(\bigcup_k E_k) = \infty$, so the equality holds in this case. A similar argument shows that if the collection is finite and contains at least one infinite set then the equality holds by virtue of both quantities being infinite. We thus only need to check the case that $\{E_k\}_{k=1}^{K}$ for some $K > 0$ with $c(E_k)$ finite.

Consider the case $K = 2$; that is the collection of two nonempty finite disjoint sets E, F. Then there are bijections $f : E \to \{1, \ldots, n\}, g : F \to \{1, \ldots, m\}$ for some $n, m > 0$, and we construct the function $h : E \sqcup F \to \{1, \ldots, n + m\}$ by

$$x \mapsto \begin{cases} f(x) & x \in E \\ n + g(x) & x \in F \end{cases}$$

Since E and F are disjoint, it is evident that this function is a bijection. We thus have by definition of c that $c(E \sqcup F) = n + m = c(E) + c(F)$. The case for a general K follows by a simple inductive argument. $\square$

Question 1.5 (Royden 2.8)**.** Let B be the set of rational numbers in the interval $[0,1]$, and let $\{I_k\}_{k=1}^n$ be a finite collection of open intervals that covers B. Prove that $\sum_{k=1}^n m^*(I_k) \geq 1$.

Proof. By countable subadditivity, it suffices to only consider the case where I_k are mutually disjoint. We also assume that each interval is finite and intersects $[0,1]$. This allows us to write $I_k = (a_k, b_k)$ for $a_1 < b_1 \leq a_2 < b_2 \leq a_3 < \cdots < b_{n-1} \leq a_n < b_n$ with $a_1 < 0, b_n > 1, b_1 > 0$ and $a_n < 1$. If $b_{k-1} < a_k$ for some k then the nonempty interval (b_{k-1}, a_k) is contained in the complement of the union of the I_k's. Furthermore, by the conditions $b_1 > 0$ and $a_n < 1$, (b_{k-1}, a_k) intersects $[0,1]$, and by density of the rational numbers, the I_k's may not cover the closed unit interval. Thus we have $b_{k-1} = a_k$ for all k, so $\bigcup_{k=1}^n I_k = (a_1, b_n) \setminus \{a_2, \ldots, a_n\}$. We then have

$$\begin{aligned}
\sum_{k=1}^n m^*(I_k) &\geq m^*\left(\bigcup_{k=1}^n I_k\right) \\
&= m^*((a_1, b_n) \setminus \{a_2, \ldots, a_n\}) \\
&= m^*((a_1, b_n) \setminus \{a_2, \ldots, a_n\}) + m^*(\{a_2, \ldots, a_n\}) \\
&\geq m^*((a_1, b_n)) \\
&\geq m^*([0,1]) \\
&= 1
\end{aligned}$$

as required. □

Question 1.6 (Royden 2.10)**.** Let A and B be bounded sets for which there is an $\alpha > 0$ such that $|a - b| \geq \alpha$ for all $a \in A, b \in B$. Prove that $m^*(A \cup B) = m^*(A) + m^*(B)$.

Proof. Countable subadditivity shows that $m^*(A \cup B) \leq m^*(A) + m^*(B)$, so we only need to show the reverse inequality. Given $\varepsilon > 0$, since A, B are bounded, we can take a cover $\{I_k\}_{k=1}^\infty$ for $A \cup B$ of open intervals such that each I_k is of length at most α and $\sum_{k=1}^\infty \ell(I_k) \leq m^*(A \cup B) + \varepsilon$. Note that if I_k intersects both A and B, then we will have some elements of A and B of distance less than α. This means that we can split up the cover $\{I_k\}_{k=1}^\infty$ into a cover $\{J_k\}$ of A and a cover $\{K_k\}$ of B, and so we get

$$m^*(A) + m^*(B) \leq \sum_k \ell(J_k) + \sum_k \ell(K_k) \leq \sum_k \ell(I_k) \leq m^*(A \cup B) + \varepsilon$$

Thus for all $\varepsilon > 0$, $m^*(A) + m^*(B) \leq m^*(A \cup B) + \varepsilon$, so with we must have $m^*(A) + m^*(B) \leq m^*(A \cup B)$. With the observation made at the beginning of the proof, we have $m^*(A \cup B) = m^*(A) + m^*(B)$. □

Question 1.7 (Royden 2.11)**.** Prove that if a σ-algebra of subsets of $\mathbb{R}$ contains intervals of the form (a, ∞), then it contains all intervals.

Proof. Let $\mathcal{A}$ be such a σ-algebra. We have the following cases to consider:

$$(a,b), [a,b], (a,b], [a,b), (-\infty, b), (-\infty, b], [a, \infty), (-\infty, \infty) \text{ and } \varnothing.$$

For the last two, we have $(-\infty, \infty) = \bigcup_{n=1}^\infty (-n, \infty) \in \mathcal{A}$ and $\varnothing = (-\infty, \infty)^c \in \mathcal{A}$. $(-\infty, b] = (b, \infty)^c \in \mathcal{A}$, and so $(a, b] = ((-\infty, a] \cup (b, \infty))^c \in \mathcal{A}$. Since $(-\infty, b]$ for all b, we also have $(-\infty, b) = \bigcup_{n=1}^\infty (-\infty, b - 1/n] \in \mathcal{A}$, which also gives us $[a, \infty) = (-\infty, a)^c \in \mathcal{A}$. We now can also get $(a, b) = ((-\infty, a] \cup [b, \infty))^c \in \mathcal{A}$, $[a, b] = ((-\infty, a) \cup (b, \infty))^c \in \mathcal{A}$, and $[a, b) = ((-\infty, a) \cup [b, \infty))^c \in \mathcal{A}$. Therefore, $\mathcal{A}$ contains all intervals. □

Question 1.8 (Royden 2.19)**.** Let E have finite outer measure. Show that if E is not measurable, then there is an open set $\mathcal{O}$ containing E that has finite outer measure and for which

$$m^*(\mathcal{O} \setminus E) > m^*(\mathcal{O}) - m^*(E).$$

Proof. Since E is not measurable, there is an $\varepsilon > 0$ such that for all open sets $\mathcal{O}$ containing E we have $m^*(\mathcal{O} \setminus E) \geq \varepsilon$. By definition of measure, we can take a collection of open intervals $\{I_k\}_{k=1}^{\infty}$ such that it covers E and $\sum_{k=1}^{\infty} \ell(I_k) < m^*(E) + \varepsilon$. Letting $\mathcal{O} = \bigcup_{k=1}^{\infty} I_k$, we see that

$$m^*(\mathcal{O}) \leq \sum_{k=1}^{\infty} \ell(I_k) < m^*(E) + \varepsilon \Longrightarrow m^*(\mathcal{O}) - m^*(E) < \varepsilon \leq m^*(\mathcal{O} \setminus E)$$

which shows that $\mathcal{O}$ has finite measure and proves the claim. $\square$

Question 1.9 (Royden 2.20: Lebesgue)**.** Let E have finite outer measure. Show that E is measurable if and only if for each open, bounded interval (a, b),

$$b - a = m^*((a, b) \cap E) + m^*((a, b) \setminus E).$$

Proof. Forward direction follows directly from the defintion of measurable sets. For the backward direction, it suffices to show for a given $\varepsilon > 0$ the existence of an open set $\mathcal{O}$ containing E with the property $m^*(\mathcal{O} \setminus E) < \varepsilon$. By definition of measure, take a cover $\{I_k\}_{k=1}^{\infty}$ of E consisting of open intervals such that $\sum_{k=1}^{\infty} \ell(I_k) < m^*(E) + \varepsilon$. Since E is bounded, we can assume that $I_k = (a_k, b_k)$ are bounded. Letting $\mathcal{O} = \bigcup_{k=1}^{\infty} I_k$ we get

$$\begin{aligned}
m^*(\mathcal{O} \setminus E) &\leq \sum_{k=1}^{\infty} m^*((a_k, b_k) \setminus E) \\
&= \sum_{k=1}^{\infty} ((b_k - a_k) - m^*((a_k, b_k) \cap E)) \\
&= \sum_{k=1}^{\infty} \ell(I_k) - \sum_{k=1}^{\infty} m^*(I_k \cap E) \\
&< m^*(E) + \varepsilon - m^*(\mathcal{O} \cap E) \\
&= \varepsilon
\end{aligned}$$

As mentioned earlier, this shows that E is measurable. $\square$

Question 1.10 (Royden 2.24)**.** Show that if E_1 and E_2 are measurable, then

$$m(E_1 \cup E_2) + m(E_1 \cap E_2) = m(E_1) + m(E_2).$$

Proof. Since they are measurable, we have $m(E_1) = m(E_1 \cap E_2) + m(E_1 \setminus E_2)$ and $m(E_2) = m(E_1 \cap E_2) + m(E_2 \setminus E_1)$. Adding these, we get

$$\begin{aligned}
m(E_1) + m(E_2) &= m(E_1 \cap E_2) + (m(E_1 \cap E_2) + m(E_1 \setminus E_2) + m(E_2 \setminus E_1)) \\
&= m(E_1 \cap E_2) + m(E_1 \cup E_2)
\end{aligned}$$

as required. $\square$

Question 1.11 (Royden 2.25)**.** Show that the assumption that $m(B_1) < \infty$ is necessary in part (ii) of the theorem regarding continuity of measure.

Proof. It is not necessary in the sense that if the result of the theorem holds then $m(B_1) < \infty$, but it is necessary in the sense that the result may not hold if we do not assume this. For example, consider the decreasing sequence $\{B_k = (k, \infty)\}$. Each of these has infinite measure so $\lim_{k\to\infty} m(B_k) = \infty$. However, $\bigcap_{k=1}^{\infty} B_k = \varnothing$ so $m(\bigcap_{k=1}^{\infty} B_k) = 0$, which contradicts the result of the theorem. $\square$

Question 1.12 (Royden 2.27)**.** Let $\mathcal{M}'$ be any σ-algebra of subsets of $\mathbb{R}$ and m' a set function on $\mathcal{M}'$ which takes values in $[0, \infty]$, is countably additive, and such that $m'(\varnothing) = 0$.

(i) Show that m' is finitely additive, monotone, countably monotone, and possesses the excision property.

(ii) Show that m' possesses the same continuity properties as Lebesgue measure.

Proof. (i). For finite additivity, let $E_1, \ldots, E_n \in \mathcal{M}'$ be a finite collection of measurable sets. We can then define $E_k = \varnothing$ for $k > n$ and apply countable additivity to get $m'(\bigcup_{k=1}^n E_k) = m'(\bigcup_{k=1}^\infty E_k) = \sum_{k=1}^\infty m'(E_k) = \sum_{k=1}^n m'(E_k)$. For excision property and monotonicity, for sets $E \subseteq F \in \mathcal{M}'$, we have $m'(F) = m'(E) + m'(F \setminus E) \Rightarrow m'(E) = m'(F) - m'(F \setminus E)$, and furthermore we have $m'(E) = m'(F) - m'(F \setminus E) \leq m'(F)$. For countable monotonicity, consider a set $E \in \mathcal{M}'$ and a cover $\{E_k\}_{k=1}^\infty \subseteq \mathcal{M}'$ of E. Construct a cover $\{F_k\}_{k=1}^\infty$ with $F_k = E_k \setminus \bigcup_{i=1}^{k-1} E_i$ and note that $F_k \subseteq E_k$, the collection is disjoint, and $\bigcup_{k=1}^\infty F_k = \bigcup_{k=1}^\infty E_k$. We thus have by countable additivity and monotonicity that

$$m'(E) \leq m'\left(\bigcup_{k=1}^\infty F_k\right) = \sum_{k=1}^\infty m'(F_k) \leq \sum_{k=1}^\infty m'(E_k)$$

Thus the four properties hold.

(ii). Let $\{E_k\}_{k=1}^\infty$ be an increasing sequence of sets in $\mathcal{M}'$. If any is of infinite measure, then both sides must be infinite by monotonicity, so continuity holds in this case. Otherwise, we construct $C_k = E_k \setminus E_{k-1}$ where we define $E_0 = \varnothing$. Since the sequence E_k is increasing, we have $\bigcup_{k=1}^\infty C_k = \bigcup_{k=1}^\infty E_k$ and the collection of C_k's is disjoint, hence countable additivity applies to give us

$$m'\left(\bigcup_{k=1}^\infty E_k\right) = m'\left(\bigcup_{k=1}^\infty C_k\right) = \sum_{k=1}^\infty m'(E_k \setminus E_{k-1})$$

We then apply the excision property to get

$$m'\left(\bigcup_{k=1}^\infty E_k\right) = \lim_{n\to\infty} \sum_{k=1}^n m'(E_k) - m'(E_{k-1}) = \lim_{n\to\infty} m'(E_n) - m'(E_0) = \lim_{n\to\infty} m'(E_n)$$

Now let $\{D_k\}_{k=1}^\infty$ be an decreasing sequence of sets in $\mathcal{M}'$ with $m'(D_1) < \infty$. We can then construct an increasing sequence $E_k = D_1 \setminus D_k$ with each E_k of finite measure by monotonicity since D_1 is such. We then have

$$\begin{aligned}
m'(D_1) - m'\left(\bigcap_{k=1}^\infty D_k\right) &= m'\left(D_1 \setminus \bigcap_{k=1}^\infty D_k\right) \\
&= m'\left(\bigcup_{k=1}^\infty D_1 \setminus D_k\right) \\
&= m'\left(\bigcup_{k=1}^\infty E_k\right) \\
&= \lim_{n\to\infty} m'(E_k) \\
&= \lim_{n\to\infty} m'(D_1) - m'(D_k)
\end{aligned}$$

Continuity of intersection of a decreasing sequence then follows. □

Question 1.13 (Royden 2.28)**.** Show that continuity of measure together with finite additivity of measure implies countable additivity of measure.

Proof. □